

Kuiyi (Cory) Gao

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Mandarin (native) | English — TOEFL iBT 110 | Cantonese (intermediate)

Education

University of North Carolina at Chapel Hill

Jan 2026 – May 2027

B.A. in Computer Science — exchange student Spring 2026, matriculated as a transfer student Aug 2026

(expected)

- Cumulative GPA 3.68/4.00.
- Advanced coursework: COMP 550 Natural Language Processing, COMP 562 Machine Learning, COMP 586 Advanced Natural Language Processing (research project), COMP 380 Technology, Ethics & Culture.

The Chinese University of Hong Kong

Sep 2023 – May 2026

B.Sc. in Computer Science, Faculty of Engineering ELITE Stream

- Advanced coursework: ELEG 5491 Introduction to Deep Learning (graduate level), ESTR 3108 Fundamentals of Artificial Intelligence (ELITE stream), CSCI 3130 Formal Languages & Automata Theory, CENG 3420 Computer Organization & Design, CSCI 2100 Data Structures, ENGG 2760/2780 Probability & Statistics for Engineers, Applied Deep Learning, Operating Systems, Database Design & Analysis, Linear Algebra.

Publications

- [1] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 (Main Conference). [\[paper\]](#) [\[code\]](#)
- [2] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [\[paper\]](#) [\[code\]](#)
- [3] **Kuiyi Gao*** et al. *A cross-platform benchmark for GUI-agent robustness under interface noise*. Under review, 2026 (*co-first author).

Research Experience

GUI-Agent Robustness under Interface Noise [3] — CUHK, advised by Prof. Michael R. Lyu

Hong Kong & remote
Sep 2025 – present

- **Co-first author.** Cross-platform benchmark (web, desktop, mobile) evaluating 7 agent frameworks on live tasks under 42 types of interface noise: overlays, dynamic content, shifted layouts, look-alike controls.
- Interface noise changes which action an agent takes and raises unsafe-action rates, not only failure rates.
- Designed the metric suite (pass@k, relative degradation, action-difference rate, unsafe-action rate, correct-prefix length) and built the mock-interface harness for computer-use, browser-use and self-operating agents.

Agent Skill Firewall [\[code\]](#) — MBZUAI UGRIP, with Dr. Steve Liu

Abu Dhabi & Beijing
Jun 2026 – present

- Built the runtime layer that checks third-party agent skills: a static pre-scan extracts declared intent, live traffic is verified against it through a canary-seeded HTTP proxy, and each PASS / DEFER / BLOCK verdict carries the evidence it came from. Four-student team; the runtime firewall is mine.
- Measured detection on stored runs of 65 malicious skills: 0 flagged, and 46 of the 65 produced no proxy-visible egress, so the detector was never reached. Recall factors as $P(\text{execute}) \times P(\text{observe} | \text{execute}) \times P(\text{detect} | \text{observed})$. Manuscript in preparation.

- **Second author.** Designed the attack methodology: a harmful query is decomposed into sub-queries and the model generates, then iteratively edits, so each step passes the filter on its own.
- Bypassed safeguards in over 60% of cases on GPT-4V, GPT-4o, Gemini 1.5 and Gemini 1.5 Pro, against 14% for prior jailbreaking methods; the Think-Twice Prompting defense proposed in the paper blocks over 95%.
- Built part of CoJ-Bench (nine safety scenarios, three editing operations, three editing elements), ran most attack and defense experiments, wrote the paper and the rebuttal.

- **Third author.** Designed the evaluation methodology and built the benchmark: 1,500 image–prompt pairs over 29 safety scenarios, with separate unsafe and over-safety subsets.
- Evaluated nine multimodal LLMs; GPT-4V misclassified 36.1% of unsafe inputs as safe and 59.9% of benign inputs as unsafe.
- Ran the vision-centric reasoning fine-tuning arm, one of three mitigation methods tested, none of which reached satisfactory performance.

- Extended the Kernel Entropic Novelty score from images to text and implemented the training scaffold. No experimental results.

Course Projects

- **Movability segmentation for embodied safety** (CUHK ELEG 5491, 2025). DeepLabV3 with a custom ASPP head over COCO-Stuff relabelled into four movability levels, as a collision-risk prior; mIoU 0.638 at 58 FPS. [\[code\]](#)
- **MemSlot: question-agnostic saliency for long-document compression** (UNC COMP 586, 2026). 32 learnable memory slots over a frozen encoder score contract spans before any question is asked; salient spans are rendered larger and compressed through an OCR vision encoder. Evaluated on CUAD against ten control arms; results pending. [\[code\]](#)
- **Objectivity of Imagist poetry, measured with text-to-image models** (CUHK ENGE3940, 2025). 20 images per poem, CLIP-embedding variance as the statistic; Imagism 0.12 vs. Romanticism 0.28 (Flux.1-dev, $p = 0.0499$).

Honors & Awards

- CUHK Faculty of Engineering ELITE Stream; Professor Joseph Lau Scholarship of Innovation; CUHK College Entrance Scholarship; CUHK exchange scholarship (UNC, Spring 2026).
- Gaokao (national college entrance examination): top 2% in Beijing, 2023.

Skills

- Python, C, Java. PyTorch, Hugging Face Transformers, Docker, Playwright, LLM APIs.